

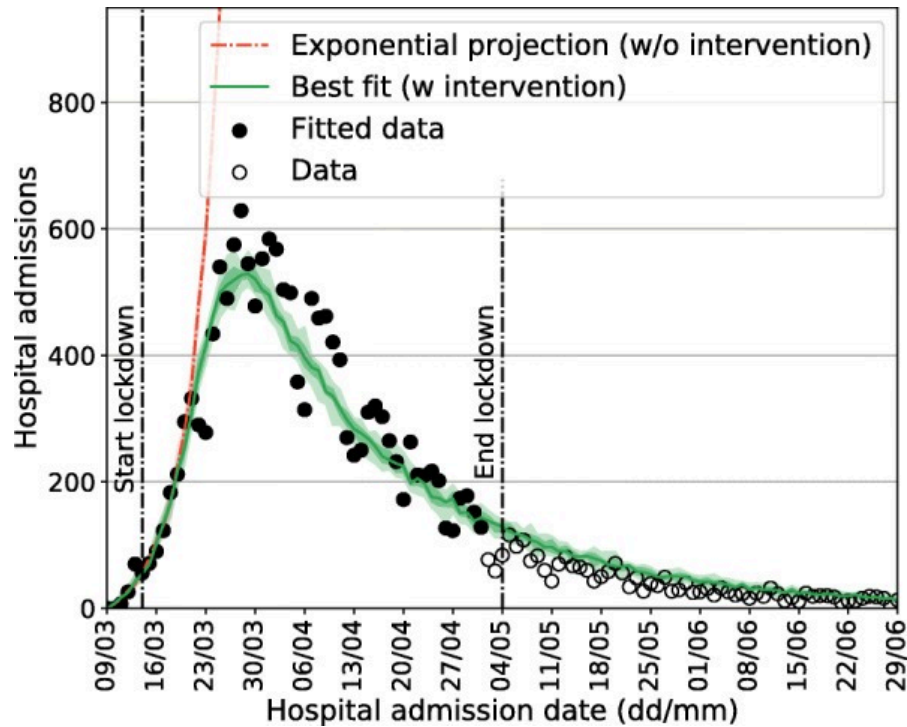
Foundations of Data Science

Lecture 1: Build, compute, critique, repeat

Prof. Gilles Louppe

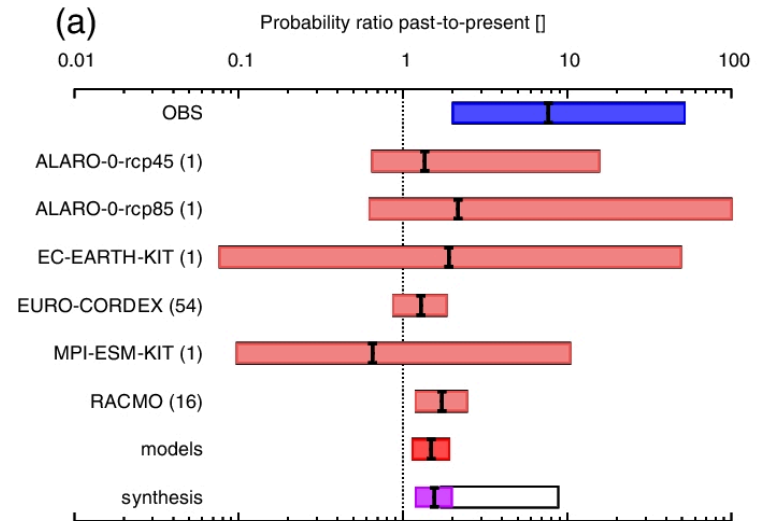
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COVID-19 in Belgium



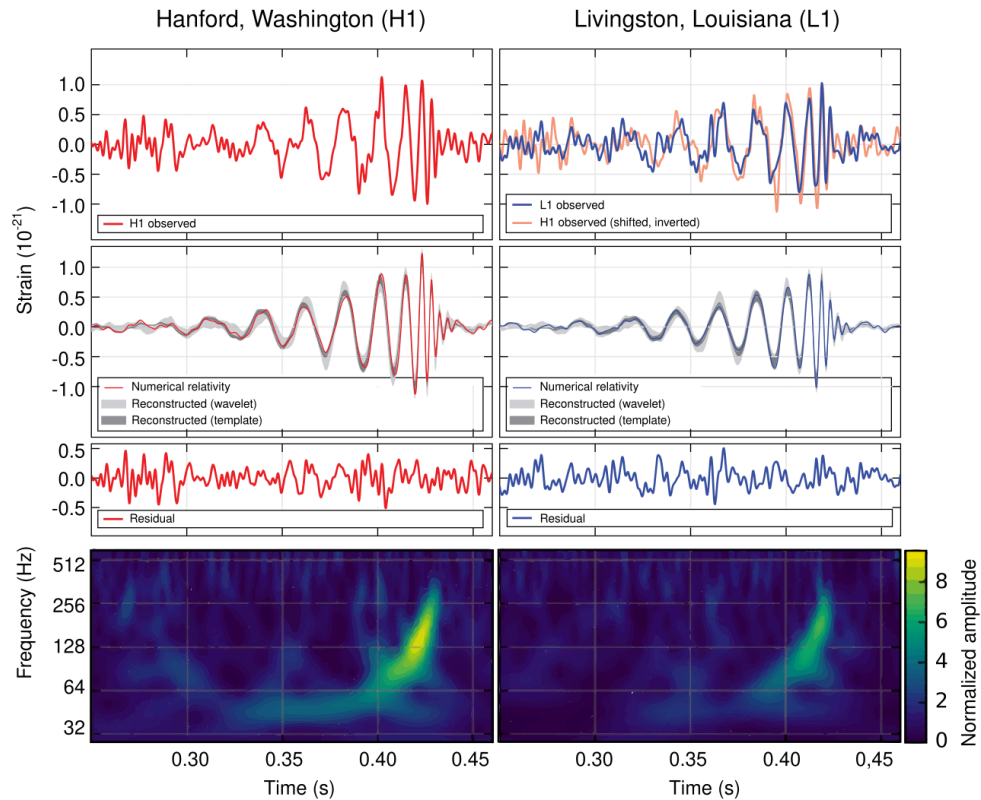
Belgian hospital admissions in 2020 (dots) and a model with lockdown (green).
Without the lockdown, the model projects exponential growth (red).

Floods in the Vesdre valley



Pepinster after the floods of July 2021, which killed 38 people in Belgium.
Climate change made such rainfall 1.2 to 9 times more likely (right).

Gravitational wave detection



First direct detection of gravitational waves, LIGO, 2015.

The measured signal matches the waveform predicted by general relativity.



Box's loop

All models are wrong, but some are useful. -- George Box.



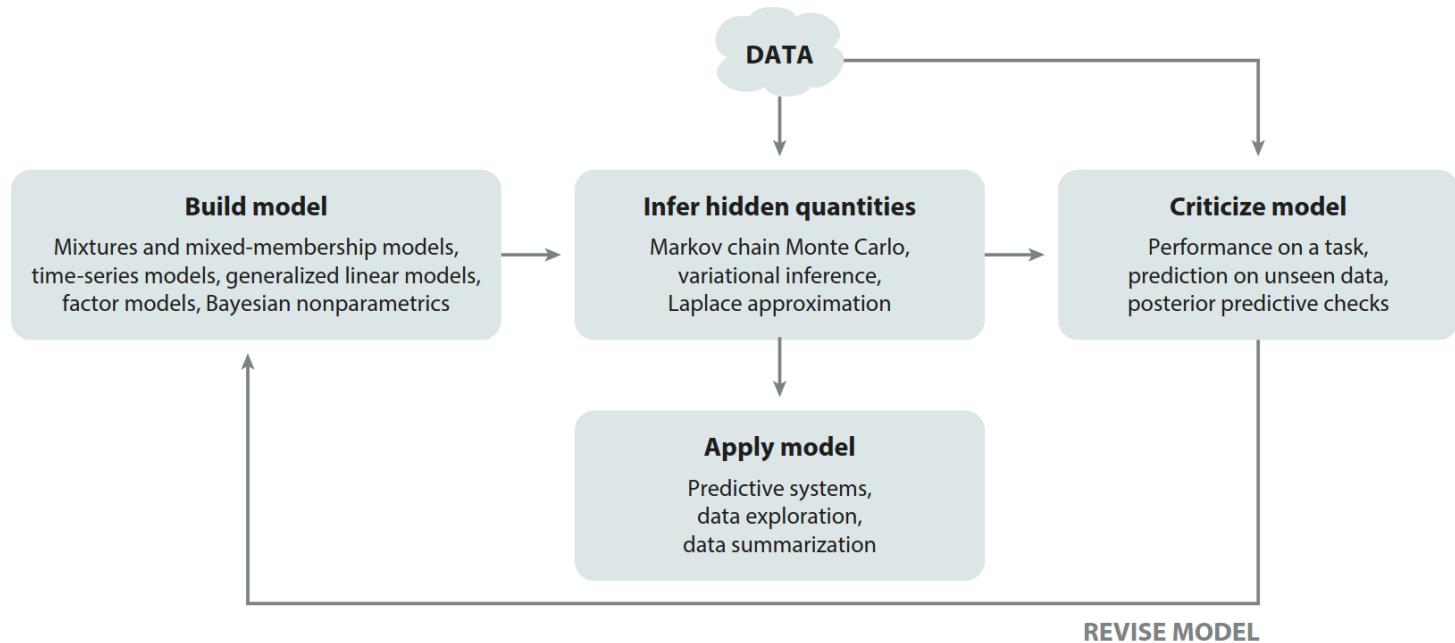
What is data science?

Data science is the discipline of **extracting knowledge** from data through the iterative application of the **scientific method**.

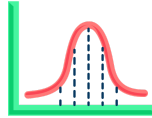
But how?



Box's loop



Scientific inquiry as an iterative process: build, compute, critique, repeat.



Step 1: Build

The first step to understanding a phenomenon is to **build a model** of it, as a simplified representation that captures its essential aspects.

- A model specifies assumptions about the data generating process.
- It encodes domain knowledge and constraints.
- It is formulated within an appropriate mathematical abstraction.
- It defines what you observe and what you do not observe but assume exists.



Step 2: Compute

The next step is to **compute** what the data tells you about the phenomenon of interest under the assumptions of your model.

- Fitting the model to data involves solving an optimization problem.
- Inference runs the model backward, from observed data to the unobserved quantities that could have produced them.
- Prediction is used to answer questions about future or unseen data.



Step 3: Critique

The third step is to **critique** the model and its predictions, to assess whether they are consistent with the data and domain knowledge.

- Compare predictions to observed data.
- Identify model limitations and mismatches.
- Reject the model if it fails to capture key aspects of the data.

Whenever you are fitting a model, we recommend constructing a scenario where it will fail. -- Gelman et al.

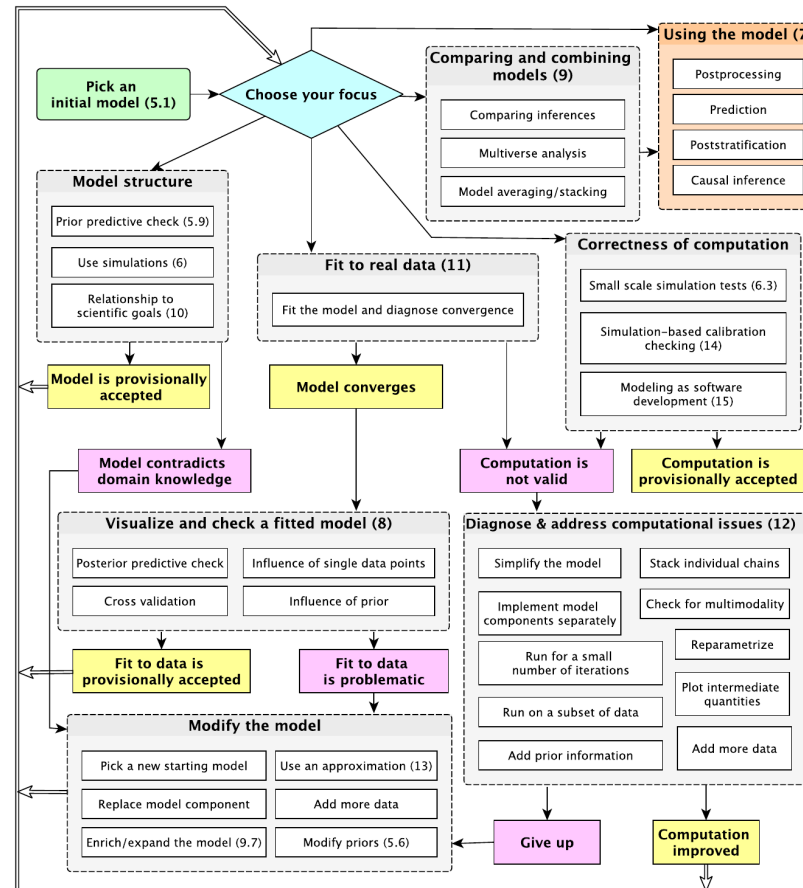


Step 4: Repeat

What you learn from the critique step informs how to **repeat** the process.

- Add complexity to the model to address its shortcomings.
- Simplify the model to improve interpretability.
- Change the model to explore alternative hypotheses.

The loop, in practice



Bayesian workflow: the steps and paths an analysis may go through.

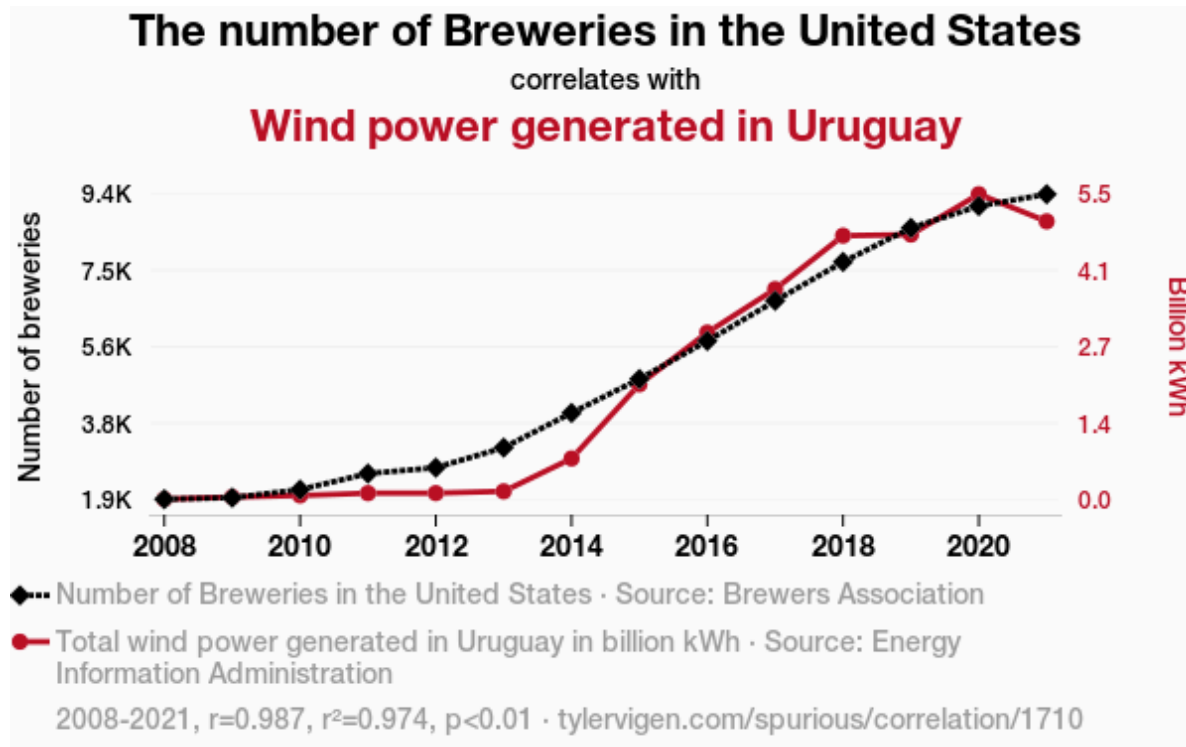
Why this approach matters?

The Fourth paradigm (Hey et al, 2009) of science emphasizes the importance of data-intensive scientific discovery. However, data alone is not enough.

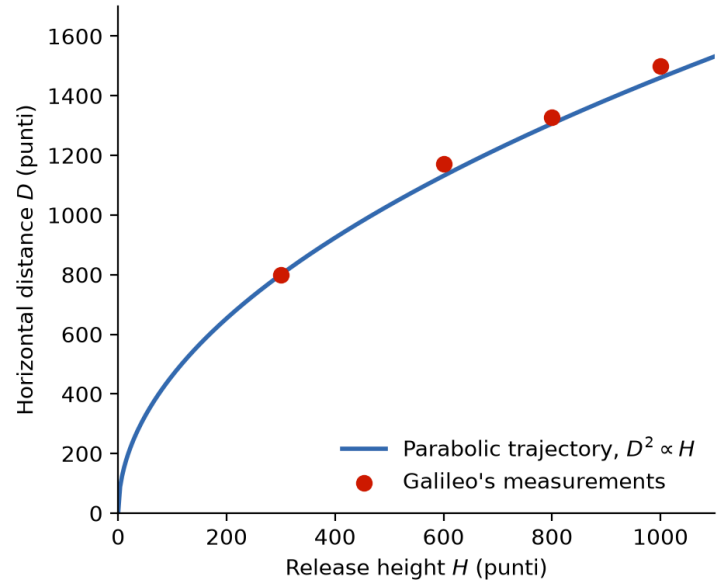
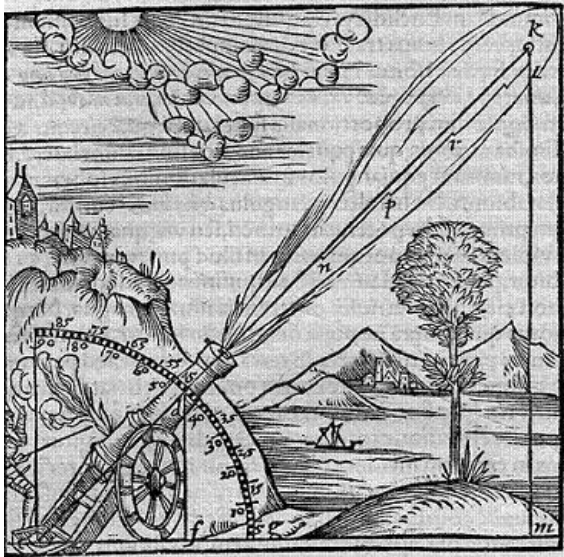
- Data without theory leads to spurious correlations.
- Theory without data leads to ungrounded speculation.
- Together, they enable **robust scientific inquiry**.

It is frequently assumed that big data can alleviate the need for careful modeling.

We do not believe this is the case. -- Gelman et al.



Spurious correlation can easily mislead data analysis.



Theory without data: the trajectory implied by Aristotelian physics (left), against Galileo's measurements, consistent with a parabola (right).

The scientific method, as embodied in Box's loop, provides a principled framework for data analysis. It contrasts with **ad-hoc data analysis practices that often lead to unreliable results**.

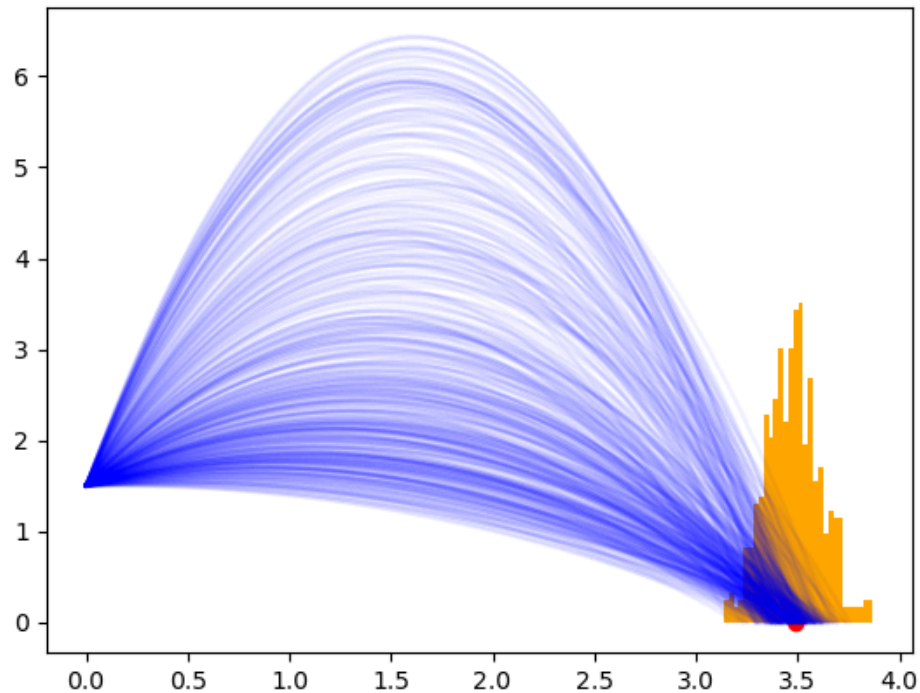
- Throw algorithms at data and see what sticks.
- Focus only on prediction accuracy.
- Treat models as black boxes.
- Stop at the first somewhat satisfactory result.

Box's loop encourages

- transparent reasoning about assumptions,
- understanding mechanisms behind data, not just correlations,
- honest assessment of model limitations,
- continuous improvement and learning.

Today's example: Projectile motion

A ball is thrown and lands at some measured distance x . What can infer about the initial velocity v and angle α of the throw?



Let's code!

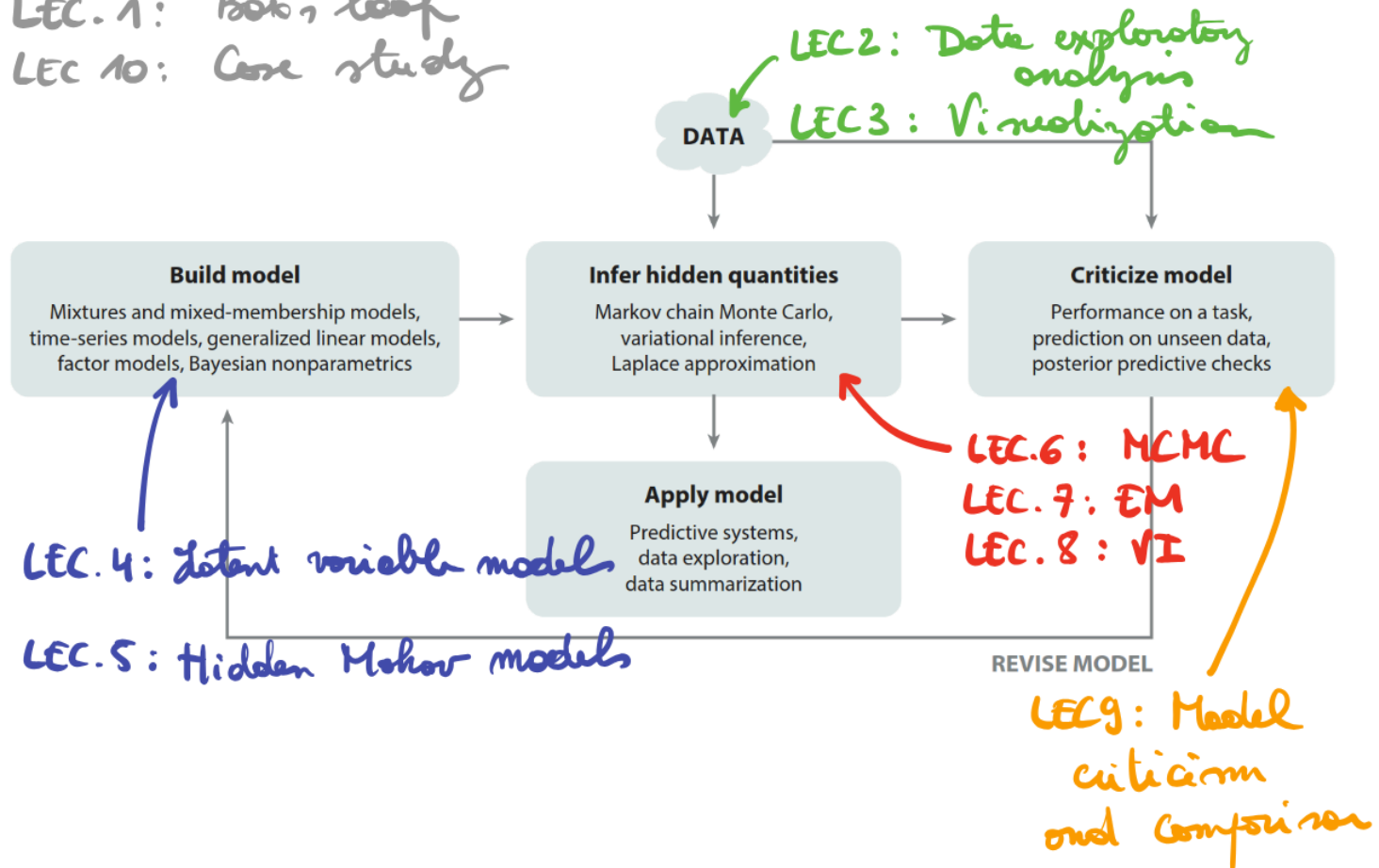
- Open a Jupyter notebook and follow along (or check `nb01.ipynb` after the lecture).
- Implement Box's loop step by step:
 1. Build a simple physical model of projectile motion.
 2. Compute estimates of v and α from data.
 3. Critique the model fit and predictions.
 4. Repeat by refining the model to account for more realistic factors.

DATS0001

Outline

- Lecture 1: Build, compute, critique, repeat
- Lecture 2: Data and exploratory analysis
- Lecture 3: Visualization
- Lecture 4: Latent variable models
- Lecture 5: State-space models
- Lecture 6: Markov Chain Monte Carlo
- Lecture 7: Model criticism and validation
- Lecture 8: Expectation-Maximization
- Lecture 9: Variational Inference
- Lecture 10: Simulation-based inference
- Lecture 11: Case study

LEC. 1: Box's loop
LEC 10: Case study



My mission

Teach you how to think like a scientist in the age of data.

By the end of this course, you will be able to explore, model, and reason about data in a principled way, but also to communicate your findings effectively.

The end.